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Neither Too Little, nor Too Late

Restructuring as a Reflex for Outcomes

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Abstract

This paper examines whether project restructuring improves World Bank project performance. Using panel data on Implementation Status and Results ratings, it combines two-way fixed effects with the PanelMatch estimator to address concerns that restructurings are endogenously timed. Restructurings consistently raise Implementation Status and Results ratings, and these gains persist across successive reporting cycles. Timing and scope both matter: early restructurings generate durable improvements, while late interventions yield shorter-lived boosts bounded by project horizons. Level I restructurings produce larger effects than Level II adjustments. These patterns show that adaptation works best when it is timely and substantive. More broadly, restructuring should be viewed not as a reactive correction but as an ordinary mechanism of adaptive management—a structured learning process that transforms performance signals into actionable design updates, reinforcing institutional flexibility and credible mid-course correction.

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Produced by the Research Support Team

Neither Too Little, nor Too Late:
Restructuring as a Reflex for Outcomes

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1 Introduction

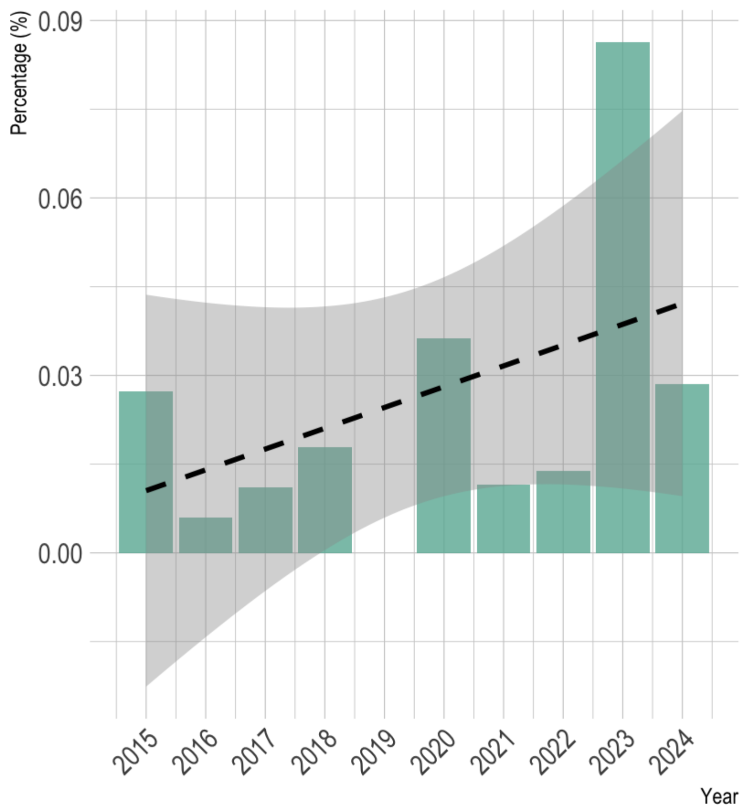
Why do some development projects succeed while others falter? Scholars and practitioners have long debated the determinants of World Bank project outcomes. (For a review, see Chasanah, Gunawan and Baroudi 2024.) Yet one dimension of project management – restructuring during implementation – remains understudied. Restructuring is not rare: the World Bank conducts more than 750 restructurings annually,[1] and references to restructuring in *Results and Performance of the World Bank Group* (RAP) reports have steadily increased over the past decade (see Figure 1). Despite this growing attention, systematic evidence on whether and how restructurings improve project performance remains scarce. This absence of evidence is puzzling: restructurings are central to adaptive management (Holling 1978; Walters 1986), allowing teams to revise objectives, reallocate resources, and recalibrate indicators in response to new information.

Do restructurings improve project performance? Does their degree of scope affect the magnitude of improvement? And at what stage of implementation are they most effective? To examine these questions, we analyze changes in *Implementation Status and Results Reports* (ISR) ratings – the Bank's primary real-time measure of project performance. We assemble a panel of World Bank projects with recorded restructuring histories and ISR ratings, employing both two-way fixed effects (TWFE) models and the PanelMatch estimator (Rauh, Kim and Imai 2025). This approach allows us to estimate dynamic treatment effects while mitigating concerns that restructurings are endogenously timed in response to deteriorating performance.

Our results show that restructured projects experience significant and persistent improvements in ISR ratings. Scope and timing both matter: Level I restructurings produce larger upgrades than Level II adjustments, and early restructurings generate durable gains that accumulate over successive reporting cycles, while late restructurings deliver more temporary boosts constrained by shorter horizons. Timing and scope thus operate as complements – early, substantive restructurings create the institutional space for learning and compounding improvement.

[1] Author's calculations based on internal data.

Figure 1: Mentions of Restructuring in RAP Reports, 2015–2024



Note: Figure 1 shows the frequency of restructuring mentions relative to total words in RAP reports, 2015–2024. The dotted line depicts the fitted linear trend, with shaded areas indicating the 95 percent confidence interval. The 2019 RAP report was not issued, and thus no data are available for that year.

We make three key contributions. First, we provide the first systematic evidence linking project restructuring to ISR-based performance trajectories, complementing prior work that focuses on final outcome ratings (ICRs) at closure (e.g., Denizer, Kaufmann and Kraay 2013; IEG 2016, 2024). Second, we show that the timing and scope of restructurings jointly determine their effectiveness, highlighting when and how adaptive management succeeds in practice. Third, we situate restructuring within broader theories of organizational learning and adaptive governance, conceptualizing it as a structured learning event – a bounded moment when teams absorb feedback, revise their theory of change, and reallocate resources. Embedding such adaptive mechanisms earlier in the project cycle can enhance both accountability and performance.

2 Determinants of World Bank Project Outcomes

What drives the success of World Bank projects is central to improving aid effectiveness. Existing literature identifies determinants operating at three levels: country-level conditions including institutional quality and macroeconomic stability; project-specific characteristics such as design and supervision; and individual-level factors, most notably the experience and incentives of task team leaders (TTLs). (For a review, see Chasanah, Gunawan and Baroudi 2024.)

A large literature emphasizes the importance of country-level conditions in determining World Bank project outcomes. Early studies find stronger performance in countries with robust civil liberties (Isham, Kaufmann and Pritchett 1997) and undistorted macroeconomic and trade policies (Kaufmann and Wang 1995; Isham and Kaufmann 1999). Building on this, Dollar and Levin (2005) and Denizer, Kaufmann and Kraay (2013) show that institutional quality – such as rule of law and bureaucratic capacity – is a key determinant of project success.

Other work highlights the risks posed by instability and political incentives. Economic instability lowers project performance (Guillaumont and Laajaj 2006), while the likelihood of success rises with the duration of peace (Chauvet, Collier and Duponchel 2010). Political distortions also matter: Dreher et al. (2013) show that projects underperform when recipient governments hold temporary UN Security Council seats while simultaneously facing economic vulnerabilities, such as high debt burdens. These findings highlight that project success depends not only on static institutional quality, but also on broader political and macroeconomic factors.

Project-level characteristics, including preparation and supervision, also shape World Bank project outcomes. Evidence on preparation is mixed. Deininger, Squire and Basu (1998) find that greater resources devoted to preparation are associated with lower project success rates, Kilby (2015) report a positive relationship between preparation duration and performance, and Dollar and Svensson (2000) find no significant effect using an instrument for staff-weeks of preparation. Evidence on supervision is similarly mixed. Kilby (2000, 2015) and Ika (2015) show a positive and significant impact of supervision on performance. Early supervision, in particular, is much more effective than later supervision. Further, Chauvet, Collier and Fuster (2017) find that supervision is more effective in improving project performance when the degree of congruence of donor-government interests is least congruent. On the other hand, Denizer, Kaufmann and Kraay (2013) show that devoting greater resources to supervision results in worse outcomes. These differences partly reflect variation in empirical strategies. Some estimates capture correlations between preparation and supervision efforts and project outcomes, while others attempt to identify causal effects by leveraging exogenous variation in the intensity of those efforts.

Finally, scholars document how individual characteristics shape project success. Denizer, Kaufmann and Kraay (2013) show that TTL fixed effects explain variation in project success comparable to country institutions, with higher-quality TTLs and less turnover yielding better outcomes. Ashton et al. (2023) refine this by estimating staff "value-added," finding that prior World Bank experience, rather than demographics, predicts success, with preparation and late-stage supervision particularly influential. Heinzel and Liese (2021) further show that supervisory ability and country experience improve recipient performance, especially in IPF projects where discretion is greater. Yet not all staff-related factors matter: Honig (2020) finds that simply co-locating World Bank staff in recipient countries has negligible impact on project success.

Taken together, existing studies show that World Bank project success depends on a mix of country conditions, project features, and staff characteristics. Good institutions and stable politics help, but strong project design and effective supervision also matter. Recent work highlights the importance of task team leaders, showing that experience and continuity improve outcomes.

3 Project Restructuring

We focus on another, less examined dimension of World Bank operations: project restructuring. World Bank projects operate in environments where political, economic, and social conditions change frequently. Even well-designed projects face unexpected challenges – such as macroeconomic shocks, leadership turnover, natural disasters, or capacity shortfalls – that can undermine implementation. To manage these risks, the Bank maintains a formal restructuring policy that enables projects to adjust during implementation. Restructuring is not simply administrative. It is intended to preserve development effectiveness by realigning project design with evolving circumstances.

The rationale for restructuring reflects a tension between commitment and flexibility. Projects must retain clear objectives, uphold fiduciary standards, and comply with social and environmental safeguards. Yet rigid adherence to initial designs can render projects ineffective when conditions change. Restructuring provides a mechanism for recalibration: it allows objectives, financing, or implementation arrangements to be revised without discarding the operation itself. In this way, the policy institutionalizes adaptability as a means of protecting development outcomes over the multi-year life of a project.

The World Bank distinguishes between two levels of restructuring under Investment Project Financing (IPF) and Program-for-Results (PforR). The classification reflects whether a change materially alters project objectives, risks, or governance arrangements.

For IPF operations, Level One restructurings apply when changes affect the project's risk profile or core contractual terms. These include upgrading the safeguard category to Category A, extending the expiration date of a Bank guarantee, relying on alternative procurement arrangements, or – under the Multi-Phase Programmatic Approach (MPA) – making a substantive change to program objectives or adding a new borrower. In such cases, approval is required from the Board, unless authority for the original financing rested with management (see Appendix B.1).

Likewise, for PforR operations, the main triggers for Level One restructurings involve changes to program development objectives, the addition of a new borrower, or the introduction of new Disbursement-Linked Indicators. As with IPF, Board approval is required unless the original financing was approved at the management level (see Appendix B.2).

All other modifications – such as revising indicators, reallocating funds, or extending loan closing dates – are treated as Level Two restructurings. These are normally approved by Bank management, though they may be escalated to the Board if the changes pose major institutional or operational risks.

Project restructurings at the World Bank follow a standardized sequence designed to balance borrower ownership with internal accountability. The process is initiated when both the Borrower and the Bank agree that material changes are necessary, typically on the basis of formal documentation such as aide memoires, management letters, or minutes of supervision meetings. The Borrower then submits a formal request, which serves as the trigger for preparation of a restructuring package (OPCS 2024).

Preparation is led by the Task Team Leader, who compiles the necessary documentation and obtains internal clearances. The package usually consists of the Borrower's request, a Restructuring Paper setting out the rationale and scope of proposed changes, and – for major revisions – a draft Memorandum of the President. Legal agreements and safeguard instruments are amended as needed, with input from specialized departments such as Legal, Finance, and Environmental and Social Standards.

Restructurings then undergo internal review and approval, with the level of scrutiny depending on the nature of the change. For "Level One" restructurings, the package is submitted to the Board of Executive Directors. "Level Two" restructurings are approved by management at the Country Director or Regional Vice President level.

Once approval is secured, the decision is formalized and communicated to the Borrower. Legal agreements are amended where necessary, and updated project documents – such as Environmental and Social Commitment Plans – are disclosed.

Beyond the procedural steps, the restructuring process reflects broader institutional tradeoffs within the World Bank. On one side is the imperative to maintain development effectiveness: restructurings are designed to ensure that projects remain aligned with country priorities and Bank policies, even as conditions shift. On the other side is the need to safeguard accountability: every restructuring involves extensive documentation, internal clearances, and disclosure requirements to prevent the process from being used to dilute standards or obscure implementation problems. This dual emphasis – on adaptability and accountability – illustrates how the Bank operationalizes flexibility within a rules-based system.

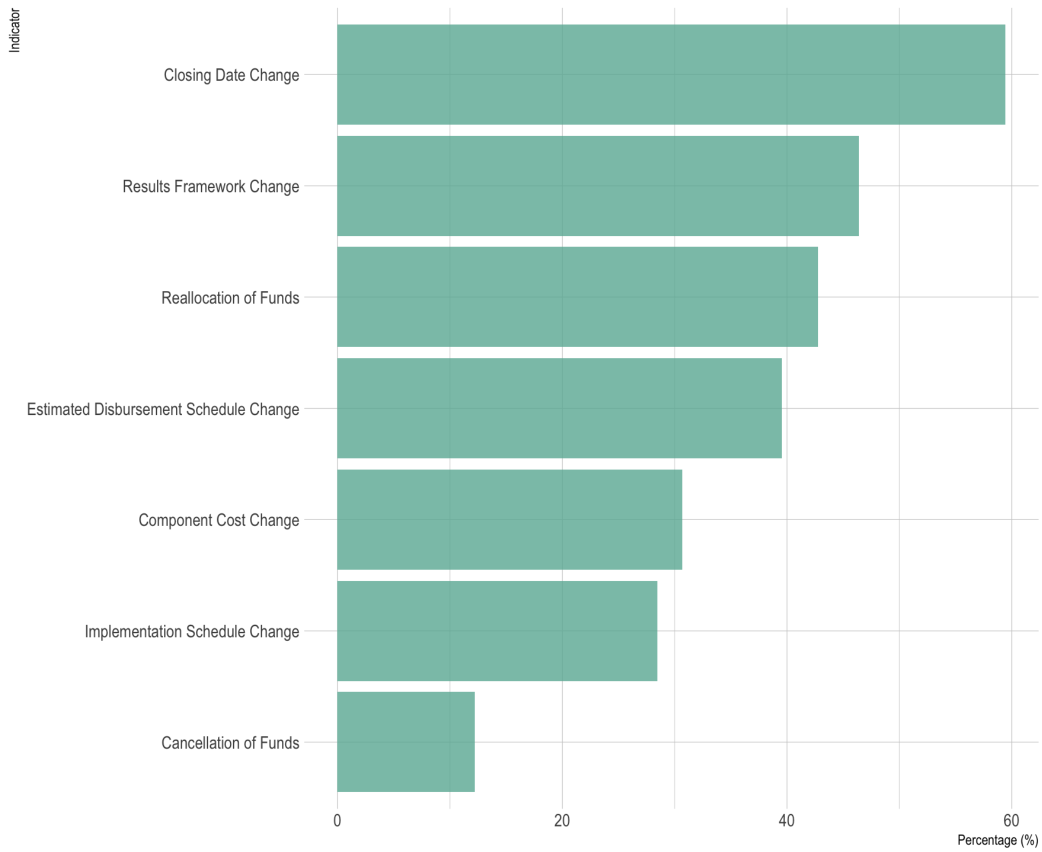
In practice, restructurings vary widely in frequency and scope. Some projects undergo only minor adjustments, such as reallocating funds across expenditure categories. Others experience multiple restructurings over their lifetime, reflecting complex implementation environments where objectives or institutional arrangements must be revised repeatedly. Patterns also differ by sector and region: infrastructure and social protection projects, for example, often confront exogenous shocks (macroeconomic crises, natural disasters, pandemics) that require formal restructuring. This variation underscores that restructuring is not an exceptional event but an integral part of portfolio management.

Figure 2 shows the relative frequency of restructuring components that account for more than 10% of all restructurings between FY2013 and FY2025.[2] A single restructuring may involve multiple components. Each bar therefore represents the share of restructurings that included a change in the corresponding component – that is, the number of restructurings with that component divided by the total number of restructurings. Closing-date and results-framework changes are the most common, appearing in roughly half of all restructurings, while cancellations occur in only about one in ten. These patterns suggest that restructurings are typically used to adjust timelines, update project logic, or realign resource flows rather than to terminate activities.

[2]The components can include: (1) a change in the Development Objectives; (2) modifications to the results framework; (3) a change in the Environmental Assessment category; (4) a change in the safeguards triggered; (5) a change in the costs associated with different components; (6) a change in the financial plan; (7) a change in the project's closing date; (8) a change in the disbursement arrangements; (9) a change in the Deferred Drawdown Option status; (10) a change in the estimated disbursement schedule; (11) a cancellation of funds; (12) a reallocation of funds; (13) a change in the risk analysis; (14) a change in the legal covenants; (15) a change in the institutional arrangements; (16) a change in the financial management arrangements; (17) a change in the procurement arrangements; (18) a change in the Advance Procurement Action; (19) a change in the implementation schedule; (20) a change in the economic and financial analysis; (21) a change in the technical aspects of the project; (22) a change in the social analysis; (23) a change in the environmental analysis; or (24) a policy waiver.

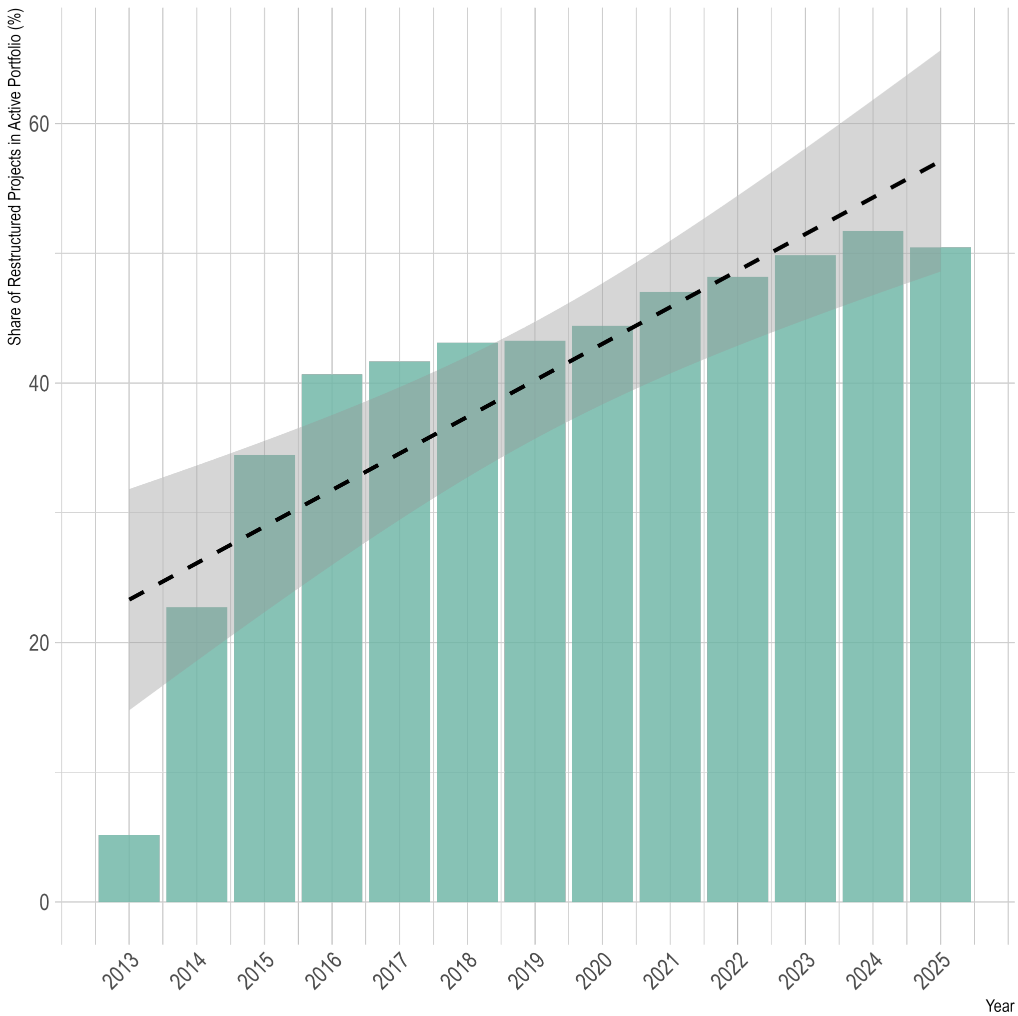
Figure 3 provides a descriptive overview of restructuring prevalence over time. It plots the share of active projects that were restructured by each financial year between FY2013 and FY2025. The share increased steadily, from around 5 percent in FY2013 to nearly 50 percent in recent years. This upward trajectory indicates that, even after accounting for portfolio expansion, an increasing portion of the Bank's active projects had already undergone at least one restructuring. The pattern likely reflects the combined effects of rising operational complexity and exposure to external shocks – such as macroeconomic volatility and the COVID-19 pandemic – which have required more mid-course adjustments. Taken together, the evidence suggests that restructuring has gradually shifted from a rare corrective measure to a routine instrument of adaptive portfolio management. Level I restructurings remain exceptionally rare – around 2% of all cases – underscoring that the overwhelming majority of portfolio adjustments (98%) occur through Level II restructurings.

Figure 2: Frequency of Restructuring Components, FY2013–FY2025



Note: Figure 2 shows the relative frequency of restructuring components that account for more than 10% of all restructurings between FY2013 and FY2025. Each bar represents the share of restructurings that involved a change in the corresponding component. The data are drawn from the World Bank's internal Databricks-hosted database.

Figure 3: Restructuring Rate by Financial Year, FY2013–FY2025



Note: Figure 3 plots the share of active projects that had restructured by or before each financial year between FY2013 and FY2025. The dashed line represents the fitted linear trend, and the shaded area denotes the 95 percent confidence interval. The restructuring data come from the World Bank's internal Databricks-hosted database.

4 Disclosable Restructuring Papers

To better understand how the World Bank adapts projects during implementation, we analyze *Disclosable Restructuring Papers*, the official documents that record and communicate mid-course adjustments.

Restructuring papers serve both informational and accountability functions. The preparation of a Restructuring Paper and its circulation through the Bank's clearance system compel project teams to reassess performance, risk management, and results frameworks. This process often reveals bottlenecks or capacity shortfalls that routine supervision may overlook, making restructuring not merely reactive but a structured mechanism for mid-course correction and organizational learning.

At the same time, the public disclosure of these papers embeds restructuring within the Bank's transparency agenda. By publishing revised objectives, financing arrangements, and safeguard commitments, the Bank ensures that changes are subject to external scrutiny rather than informal negotiation. This dual role – internal reassessment and external accountability – illustrates why restructuring is a central instrument for sustaining both the effectiveness and the legitimacy of the World Bank's development portfolio.

We assembled the corpus of these papers using the World Bank's API. Automated queries retrieved all entries available up to the most recent date at the time of writing. The API listed 5,550 papers, of which 5,545 were successfully downloaded. Each document was obtained in PDF format and converted into plain text for analysis. The earliest available restructuring paper dates to December 16, 2017, corresponding to the Second Critical Ecosystem Partnership Fund Project (P100198).

To analyze the content of restructuring papers, we construct co-occurrence networks centered on the term *restructuring*. A co-occurrence network is a graph-based representation in which words are treated as nodes and edges are formed when two terms appear within a defined textual window. The thickness of the edges corresponds to the frequency of co-occurrence, with thicker lines indicating stronger associations.

To capture not only direct pairings but also more extended relationships, we build a third-order co-occurrence network. In this structure, nodes include all words that appear within three steps of restructuring: first-order neighbors are words that directly co-occur with restructuring, second-order neighbors are words connected to those first-order terms, and third-order neighbors extend the network one step further. This design balances local semantic precision with a broader mapping of the conceptual space in which restructuring is discussed.

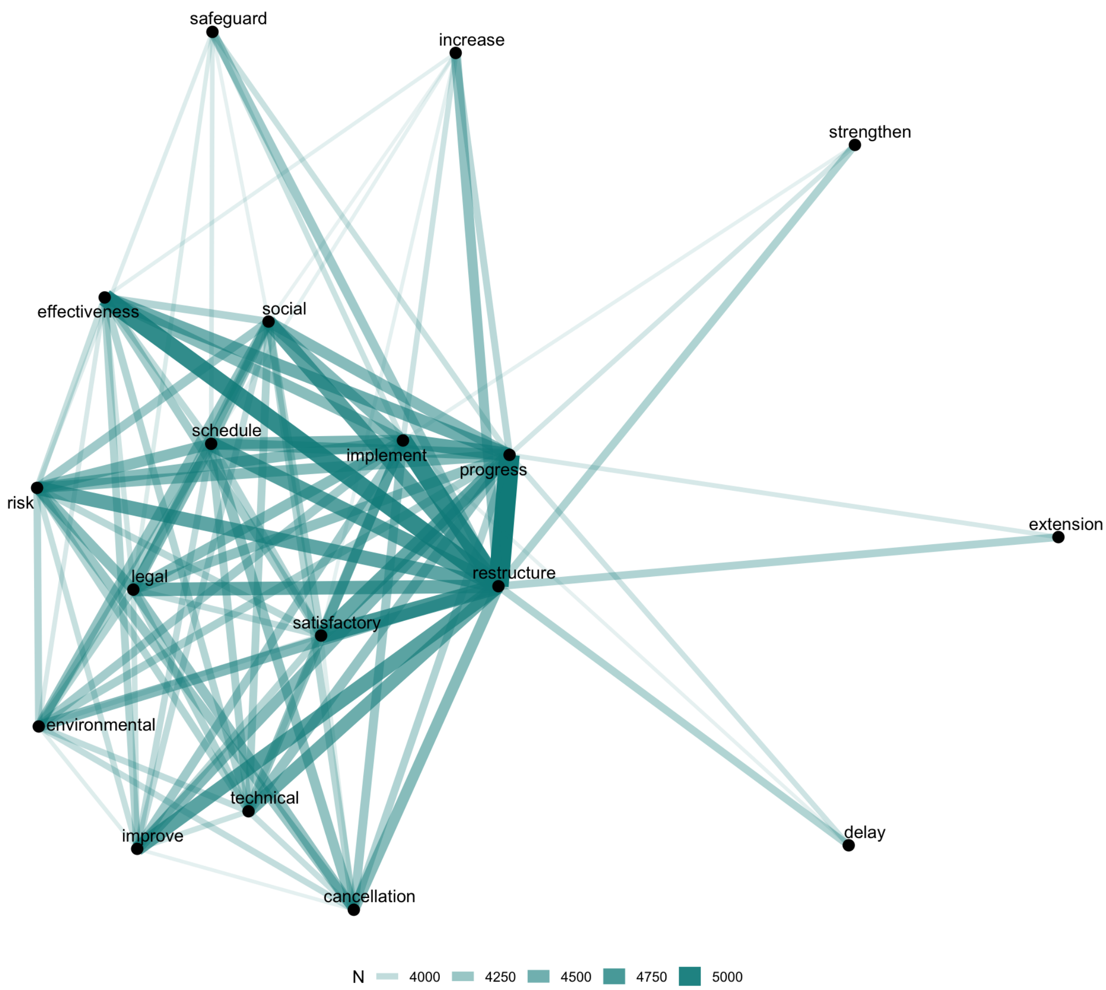
Prior to network construction, the texts undergo standard preprocessing. All words are converted to lowercase, tokenized, and stripped of common stopwords. Beyond the standard list, we implement a customized stopword dictionary to remove institution-specific boilerplate (see Appendix C). We also apply targeted lemmatization to ensure consistency, replacing *restructure* with *restructuring*. These steps create a clean corpus from which third-order co-occurrence structures can be reliably inferred.

Figure 4 shows the co-occurrence network of *Disclosable Restructuring Papers*. Several key patterns emerge. First, the strongest associations with restructuring are with implementation- and performance-related terms. The thickest edges cluster around the terms "progress," "implement," and "schedule." This suggests that discussions of restructuring are tightly bound to progress reporting and implementation details. The co-occurrence with "schedule" signals that restructuring is often triggered by slippages in planned activities, while "implement" reflects the operational focus of restructuring packages. In practice, this pattern aligns with the Bank's mid-course adjustments, where extensions or modifications are justified primarily in terms of maintaining implementation progress.

Second, the periphery shows policy-safeguard and timing concerns. Nodes such as "safeguard," "environmental," "social," and "risk" are connected but somewhat more peripheral, reflecting their importance but less frequent linkage to the operational core. This suggests that while safeguard compliance remains a necessary condition for restructuring, it is invoked episodically rather than systematically. On the opposite side, words like "delay," "extension," and "cancellation" appear at the margins, capturing outcomes when restructuring either cannot resolve bottlenecks or requires formal adjustment of timelines. This positioning reflects how restructuring acts as an intermediate mechanism – mitigating risks before outright project failure or closure.

Third, the bridging terms emphasize the Bank's balancing of technical, legal, and evaluative dimensions. "Legal," "technical," "effectiveness," and "satisfactory" serve as connectors linking the core ("progress" and "implementation") with peripheral clusters ("safeguards" and "delays"). Their presence underscores that restructuring is not purely managerial but must clear multiple evaluative and fiduciary thresholds: legal compliance, technical soundness, and demonstration of satisfactory progress toward objectives. The prominence of "effectiveness" in particular suggests that restructurings are framed not only as procedural adjustments but as instruments to preserve development effectiveness over time.

Figure 4: Co-occurrence Network of Disclosable Restructuring Papers



Note: Figure 4 only shows pairs of words with frequency greater than 3,750. Restructuring papers were retrieved using the World Bank's API. The project and restructuring data come from the World Bank's internal Databricks-hosted database.

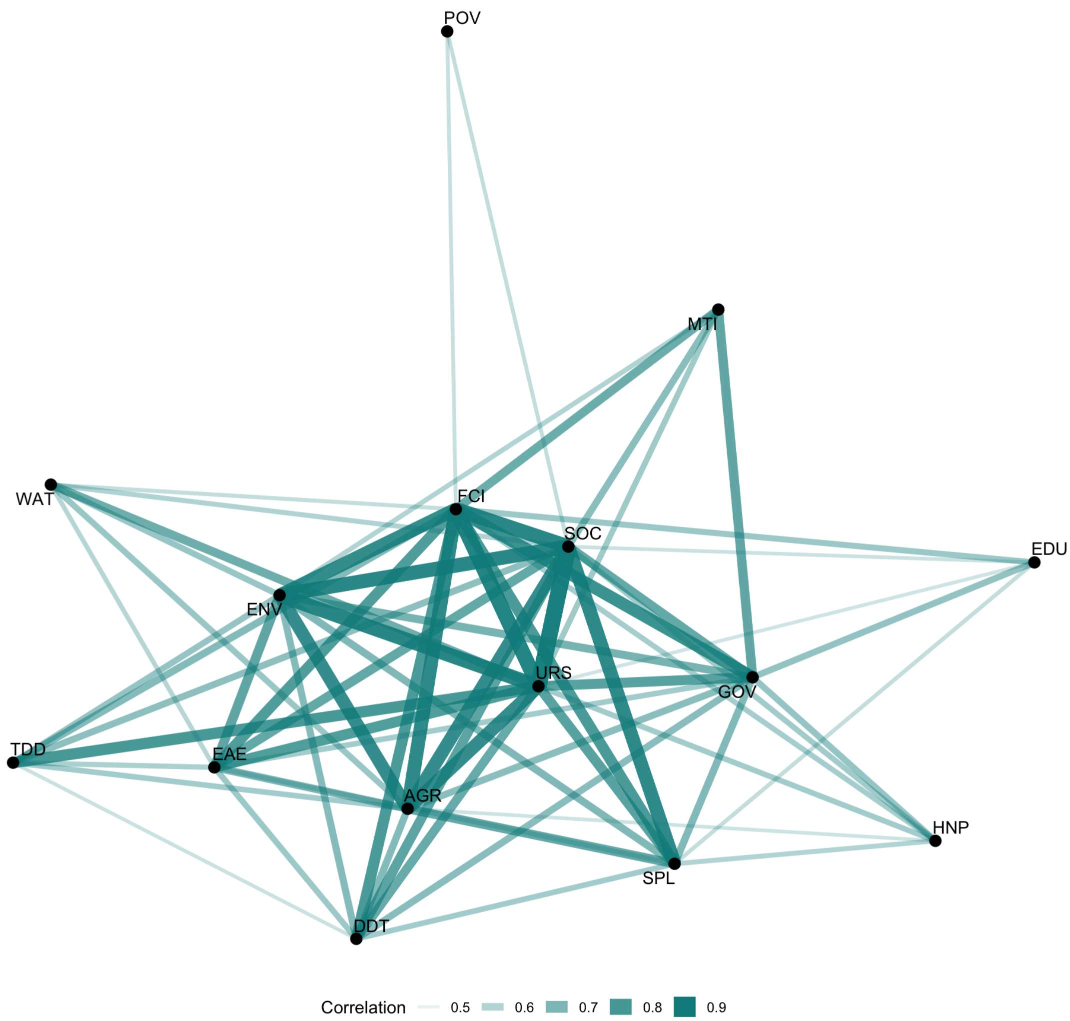
Next, we examine whether systematic patterns exist across lead Global Departments (GDs) and among borrowing countries using cosine similarity, a standard measure of textual proximity. Each corpus of restructuring papers is represented as a vector of term frequencies, and similarity is measured as the cosine of the angle between two such vectors. Values range from 0 to 1, where 0 indicates no shared vocabulary and 1 indicates identical distributions. Substantively, higher cosine similarity indicates that two groups of restructuring papers emphasize a comparable mix of concepts, while lower values suggest divergent vocabularies and thus differences in how restructurings are framed. This approach allows us to assess whether the language of project adaptation is broadly homogeneous across the Bank's portfolio or varies systematically with institutional and geographic context.

We begin by comparing similarities across GDs (see Figure 5) before turning to cross-country variation. First, a dense central cluster links practices such as Agriculture (AGR), Governance (GOV), and Finance, Competitiveness, and Innovation (FCI). Their similarity reflects a shared lexicon of safeguards, technical adjustments, and risk management typical of infrastructure-oriented operations.

Second, peripheral practices display more distinct vocabularies. The lowest similarity is between Education (EDU) and Public-Private Partnerships (PPP) (0.17), suggesting divergence between human-capital and infrastructure-financing sectors.[3] Similarly, Poverty (POV) and Water (WAT) sit at the margins, reflecting more specialized language tied to sector-specific outcomes.

[3] Because the cosine similarity falls below the visualization threshold, the corresponding nodes and edge are omitted from Figure 4.

Figure 5: Cosine Similarity Network of Disclosable Restructuring Papers, by GD



Note: Figure 5 only shows connections with a correlation greater than 0.5. Restructuring papers were retrieved using the World Bank's API. The project and restructuring data come from the World Bank's internal Databricks-hosted database.

Figure 6 shows the cosine similarity by country. First, the densest hub consists of South Asian countries, major Sub-Saharan African borrowers, and large Latin American economies. India, Pakistan, and Bangladesh occupy central positions, with strong ties to Nigeria, Ethiopia, Tanzania, and Kenya, as well as to Argentina, Brazil, and Colombia. This clustering indicates high lexical similarity across restructurings. One plausible interpretation is that these countries share a restructuring vocabulary shaped by overlapping concerns with poverty reduction, human capital, and fiscal adjustment.

Second, several middle-income reformers appear to bridge across regions. Indonesia, Morocco, Tunisia, and Sri Lanka show close connections to both South Asian and African clusters, while the Kyrgyz Republic and Mongolia link Central Asia to the broader network. These bridging positions may reflect convergence in restructuring language around governance, social protection, and infrastructure reform.

Finally, a set of peripheral or weakly connected countries stand apart. Small island states such as the Comoros and Papua New Guinea, as well as fragile or conflict-affected contexts including the Central African Republic, Mauritania, and Djibouti, are only thinly connected to the main cluster. This pattern may suggest vocabularies that are more localized or sector-specific. By contrast, upper-middle-income countries such as China, Viet Nam, and Türkiye also lie on the margins, which could indicate distinct restructuring priorities shaped by sectoral reforms and large-scale investment agendas.

Figure 6: Cosine Similarity Network of Disclosable Restructuring Papers, by Country

Note: Figure 6 only shows connections with a correlation greater than 0.7. Restructuring papers were retrieved using the World Bank's API. The project and restructuring data come from the World Bank's internal Databricks-hosted database.

5 Restructuring and Project Performance

In this paper, we argue that project restructurings function as a tool of adaptive management that directly improves project performance. Moreover, the timing of restructuring is critical: early interventions generate sustained improvements in ratings, as they mitigate drift and prevent performance erosion before institutional or political lock-in sets in.

We focus on *Implementation Status and Results Reports* (ISR) ratings, the World Bank's primary instrument for tracking project performance during implementation. Specifically, we examine two dimensions: Implementation Progress (IP) and Project Development Objectives (PDO). These scores are generated by task teams but subject to multiple layers of review by sector managers and regional leadership. Although interim in nature, ISR ratings are central to the Bank's operational apparatus. They shape the allocation of supervision resources, inform portfolio monitoring, and underpin internal accountability systems. At the same time, ISR scores carry external significance: they influence donor perceptions of credibility and signal to client governments the Bank's assessment of feasibility and risk.

Existing studies, however, have not examined how restructurings affect ISR ratings themselves. Instead, prior work focuses on final outcome evaluations at project closure. For example, Denizer, Kaufmann and Kraay (2013) find that projects restructured early in their life cycles are significantly more likely to receive satisfactory final ratings. The association is strongest among projects that exhibited early warning signs and subsequently restructured in response. Internal reviews of World Bank operations highlight a similar pattern. For instance, the *2023 Results and Performance Report* (RAP) emphasizes that timely course correction – especially before the midpoint of disbursement – is strongly associated with higher efficacy ratings at closure. Projects that restructured objectives, components, or results frameworks prior to reaching 50 percent disbursement were significantly more likely to meet their stated targets (IEG 2024). These findings suggest that while restructurings are linked to improved final ratings, little is known about the intermediate effects on ISR ratings.

We posit that ISR ratings represent the proximate channel through which restructuring effects are most likely to appear. As real-time indicators of progress and feasibility, ISR scores respond to operational adjustments, course corrections, and perceived credibility of implementation strategies. To the extent that restructuring resets project baselines, reallocates resources, or improves monitoring clarity, ISR ratings are likely to reflect those changes – especially in the near term.

The theoretical logic linking restructuring to ISR ratings rests on two features of the restructuring process. First, restructurings frequently involve revisions to project targets, indicators, or monitoring arrangements. These adjustments reduce ambiguity in supervision and improve the relevance and tractability of performance metrics. ISR scores are responsive to these changes because they track implementation alignment with stated outputs and outcomes. When indicators are refined or baseline data is clarified, the likelihood of upward ISR revisions increases – not because standards are lowered, but because progress is more accurately captured and attributable.

Second, restructuring often involves resource reallocation, including the cancellation of non-performing components or the introduction of new financing streams. These shifts affect the likelihood that planned activities will be executed within the project timeline and budget envelope. ISR ratings capture such feasibility assessments and are likely to reflect improved delivery probabilities following a well-targeted restructuring.

The marginal effect of restructuring on ISR ratings is moderated by the scope or degree of restructuring. Level I restructurings, which require Board approval, typically involve comprehensive revisions to project objectives, institutional arrangements, or financing structures. Level II restructurings, by contrast, focus on narrower operational adjustments such as reallocation of funds, modification of indicators, or minor procedural changes. From an adaptive management perspective, Level I restructurings are expected to yield larger and more persistent improvements in ISR ratings because they address multiple interlocking constraints and reset the project's theory of change. Level II restructurings, while more frequent and administratively lighter, are less likely to produce sustained performance gains, as they rarely alter the underlying causal logic of implementation. The magnitude and durability of ISR rating improvements should therefore increase with the depth of adaptation embedded in the restructuring process.

A further dimension concerns the timing. Several RAP reports find that restructurings initiated early – prior to the midpoint of disbursement or within the first half of project duration

– are more strongly associated with improved final efficacy ratings and internal assessments (IEG 2016, 2018, 2022, 2024). Multiple reasons may underlie this pattern. Early restructurings occur when implementation structures are still flexible, and before organizational routines or stakeholder expectations become fixed. Design flaws or operational mismatches can be addressed without incurring significant political or reputational costs. Moreover, the time remaining allows restructured components to be implemented and evaluated within the original timeframe. In contrast, late restructurings face compressed timelines, sunk political capital, and hardened contractual relationships. Adjustments made under these conditions may be viewed as reactive, cosmetic, or insufficient to change the overall trajectory of a project. These differences are likely to be visible in ISR ratings, which are sensitive to perceived credibility and forward momentum.

We derive three testable hypotheses from this framework:

H1: *Projects that undergo restructuring will exhibit greater improvements in subsequent ISR ratings, relative to otherwise similar projects that do not restructure.*

H2: *Projects that undergo Level I restructurings will exhibit larger improvements in ISR ratings than those undergoing Level II restructurings.*

H3: *Projects restructured earlier in the implementation cycle yield more sustained improvements in ISR ratings than those restructured later.*

6 Empirical Strategy and Results

As a preliminary test, we estimate a two-way fixed effects (TWFE) model using panel data from closed World Bank projects. This naive specification provides a baseline estimate of the association between restructuring and subsequent changes in ISR ratings. The outcome variable is the ISR rating. We focus on two dimensions: Implementation Progress (IP) and Project Development Objectives (PDO).

Our main explanatory variable is a binary indicator equal to one if the project has experienced at least one formal restructuring by the time of the given ISR report. The treatment variable is forward-looking: we compare ISR ratings before and after the first restructuring, relative to projects that have not yet restructured at the same point in their implementation. This allows us to estimate the within-project change in ratings associated with restructuring, controlling for secular trends and time-invariant differences across projects.

We estimate the following specification:

$$\text{ISR Rating}_{i,t} = \beta_0 + \beta_1 \text{Restructured}_{i,t} + \alpha_i + \delta_f + \epsilon_{i,t} \tag{1}$$

where ISR $\text{Rating}_{i,t}$ is the IP or PDO rating for project i in ISR date t, and $\text{Restructured}_{i,t}$ is an indicator equal to 1 if the project had already undergone at least one restructuring by t. The model includes project fixed effects (α_i) to absorb all time-invariant characteristics – such as project sector, size, region, implementing agency, or initial design quality – and fiscal year fixed effects (δ_f) to control for global shocks, changes in Bank-wide rating practices, or time-specific operational shifts. We cluster standard errors at the project level to allow for arbitrary serial correlation over time within projects.

The panel draws on internal Databricks-hosted databases and includes all closed investment projects with available ISR ratings and documented restructuring histories. To focus on adaptive changes rather than new financing rounds, restructurings linked to Additional Financing operations are excluded. Our empirical approach leverages within-project variation over time to identify the relationship between restructuring and performance updates, holding constant all time-invariant project characteristics and common temporal shocks. In this specification, the coefficient β_1 represents the average change in ISR ratings that follows a restructuring, conditional on baseline differences and secular trends. Consistent with Hypothesis 1, we expect β_1 to be positive.

Table 1 shows the TWFE estimates. Consistent with Hypothesis 1, restructuring is positively and significantly associated with changes in both dimensions of project performance. Following a restructuring, projects experience an average increase of 0.134 points in IP ratings and 0.059 points in PDO ratings.

Given that ISR ratings are coded on a six-point ordinal scale ranging from 1 ("Highly Unsatisfactory") to 6 ("Highly Satisfactory"), a 0.134-point increase in IP corresponds to more than one-eighth of a full-category shift. This magnitude is non-trivial, particularly in the context of ratings that tend to cluster around the midpoint of the scale. The effect on PDO, while smaller, remains substantively meaningful, especially given the higher persistence typically associated with outcome ratings.[4]

Table 1: Effect of Restructuring on ISR Ratings: Two-Way Fixed Effects Estimates

	IP	PDO
Restructured	0.134***	0.059***
	(0.015)	(0.015)
Observations	52069	57160
R^2	0.449	0.475
Adjusted R^2	0.382	0.411

Note: Clustered standard errors in parentheses. *p<0.1; **p<0.05; ***p<0.01.

While the fixed effects model provides a useful baseline, it may generate biased estimates if the timing of restructuring is endogenously determined – i.e., if restructurings are systematically implemented in response to unobserved shocks to project performance. To address this concern, we implement a matching estimator designed for panel data with time-varying treatments: the PanelMatch framework developed by Rauh, Kim and Imai (2025).

PanelMatch extends the logic of matching to longitudinal settings where treatment status varies over time. Rather than comparing all treated and untreated observations, the method constructs a matched comparison set for each treated unit based on a specified history of pre-treatment covariates, including lagged outcomes. This reduces bias from time-varying confounding and ensures that treated and control units are comparable in terms of recent performance trajectories and covariate histories prior to treatment. The estimator then compares the treated units' outcomes with those of their matched controls in the periods following treatment, allowing for dynamic treatment effects and flexible event windows.

This approach offers several advantages over standard regression or difference-in-differences estimators. First, it relaxes the parallel trends assumption by conditioning on rich pre-treatment histories, including lagged ISR ratings and other project characteristics. Second, it avoids extrap-

[4]We also report standardized coefficient estimates. The binary indicator for restructuring is associated with a 0.07 standard deviation increase in IP and a 0.03 standard deviation increase in PDO.

olation across dissimilar projects by discarding unmatched observations. Third, it provides flexible tools for assessing treatment effects over multiple leads, enabling us to examine how the impact of restructuring unfolds over time.

Given that restructuring is often triggered by deteriorating performance or management judgment, the key concern is selection on unobservables that evolve over time. By explicitly matching on pre-treatment trajectories, PanelMatch helps mitigate this concern. In what follows, we use Mahalanobis distance matching on a four-period lag history of pre-treatment variables, including prior ISR scores and project-level covariates, and estimate average treatment effects across multiple post-restructuring periods.

We focus on post-treatment dynamics over leads 0–6, corresponding to approximately three years after restructuring, given that ISR ratings are updated semiannually and the median project lifecycle in our sample is 5.6 years ($\approx$ 11–12 ISR periods). This horizon is long enough to capture the medium-run operational adjustments that restructurings are designed to trigger, while avoiding the severe right-censoring, attrition, and contamination from subsequent restructurings that arise at longer horizons. In other words, the 0–6 window provides a balance between statistical power and estimand purity, ensuring that effect estimates are not driven by the dwindling subset of unusually long-lived projects.

In constructing matched sets, we condition primarily on the log of project commitment amounts. Ideally, one would also match on additional covariates such as leading Global Department (GD), region, country, or team task leaders (TTLs). In practice, however, the implementation of factor variables with sparse support across project-year panels leads to non-invertible contrast matrices in the estimation routine. This arises because PanelMatch requires non-degenerate pre-treatment histories across all matched strata, and categories with limited or single-project representation violate this condition. Restricting the matching set to log commitments thus reflects a pragmatic tradeoff: it avoids unstable or ill-posed matches while retaining a covariate that is highly predictive of project design, scale, and subsequent performance.

Figure 7 reports dynamic treatment effect estimates from PanelMatch. The results show that restructurings are followed by consistent and positive improvements in both ISR dimensions. For

IP, the estimated effect is around 0.14 points at the time of restructuring (lead 0) and rises steadily to roughly 0.25–0.30 points within three years (lead 6). For PDO, the effect is slightly larger at onset and displays a similar upward trajectory, reaching 0.35 points at lead 6. In both cases, confidence intervals exclude zero across nearly the entire horizon, indicating robust gains.

These dynamic estimates reinforce the baseline TWFE results while addressing potential endogeneity in the timing of restructurings. The upward trend is substantively consistent with our theoretical argument: restructurings recalibrate project targets and reallocate resources in ways that generate sustained improvements in implementation feasibility and outcome alignment. Importantly, the accumulation of positive effects over successive ISR cycles suggests that restructurings are not merely cosmetic adjustments producing temporary bumps in ratings, but instead have persistent impacts on the way projects are supervised and assessed.

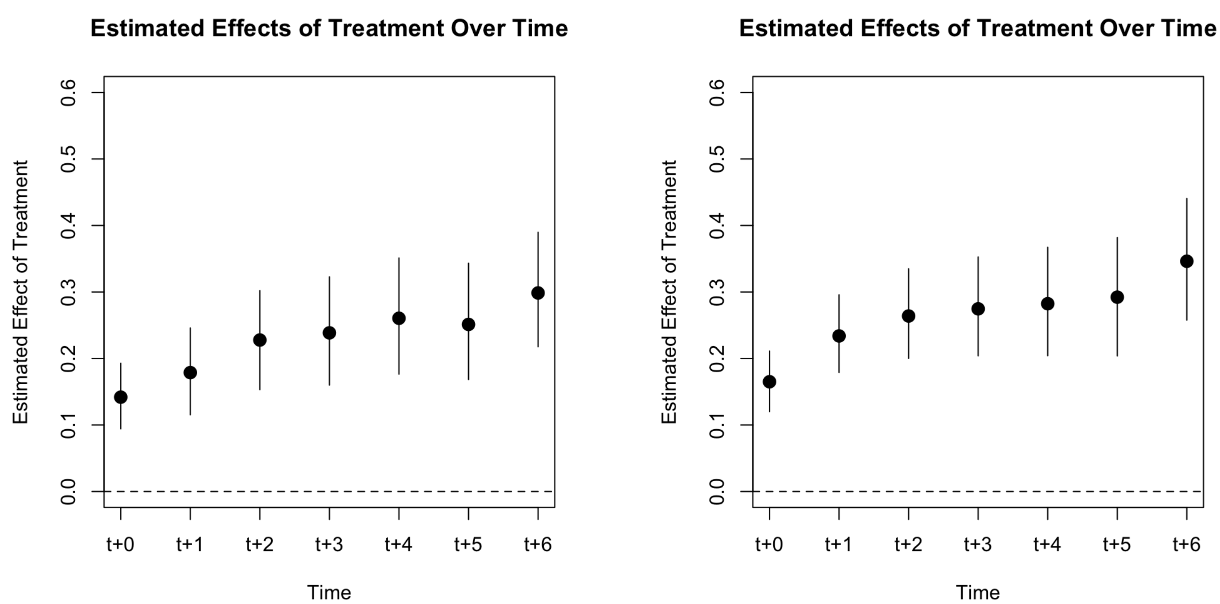
We interpret these gains not as a mechanical artifact of re-labeling, but as the product of a behavioral and institutional adjustment process. A formal restructuring does three things at once. First, it re-coordinates attention: once a team proposes a restructuring, the project is forced back onto the managerial agenda of task team leaders, technical specialists, country management, and – where relevant – the Board. That renewed attention is not cosmetic. It often results in clearer accountability, explicit sequencing of actions, and a narrowing of scope to what can realistically be delivered within remaining time and financing.

Second, restructuring realigns incentives. Before restructuring, teams can be trapped by sunk-cost bias or anchoring on original targets even when those targets have become infeasible. The act of restructuring legitimizes course correction: it converts what would otherwise look like backtracking into an institutionally authorized "fix," reducing the reputational cost of admitting that part of the design is underperforming.

Third, restructuring provides a psychological reset for both the Bank and the borrower. By re-stating what "success" means, teams restore internal and external confidence that the (revised) objective is deliverable. This credibility channel matters because ISR ratings capture not just observed progress but assessed feasibility going forward, and feasibility improves when both sides believe the plan is once again realistic.

This interpretation links restructuring to a broader class of adaptive behaviors under uncertainty. Development projects operate in volatile political and macroeconomic environments, and the original Results Framework is almost never a sufficient statistic for what ultimately gets implemented. Left unattended, teams can drift: activities slip; bottlenecks accumulate; monitoring indicators become decoupled from what is actually happening on the ground. A restructuring is therefore not simply a bureaucratic edit. It is an institutionalized interruption of drift. In that interruption, several well-documented behavioral frictions are directly confronted. Anchoring on initial disbursement plans is replaced by an updated resource map. Sunk-cost aversion is partially neutralized by management approval to cancel or downscale weak components. Over-optimism is checked by explicit risk re-assessment embedded in the Restructuring Paper. In that sense, restructuring is best understood as both a governance instrument and a cognitive aid: it forces managers to convert diffuse operational discomfort into a concrete, monitorable adaptation plan. Our empirical findings are consistent with this mechanism.

Figure 7: Dynamic Effects of Restructuring on ISR Ratings: PanelMatch Estimates



(a) IP

(b) PDO

After establishing the overall effect of restructuring on project performance, we next investigate whether different types of restructurings – Level I versus Level II – produce distinct impacts on ISR outcomes (Hypothesis 2).

Table 2 shows that Level I restructurings, which involve formal Board approval and typically signal major project redesigns, are associated with substantially larger improvements in both Implementation Progress (IP) and Project Development Objective (PDO) ratings compared to Level II restructurings. The two-way fixed effects estimates suggest an average gain of roughly 0.29 points in IP ratings and 0.21 points in PDO ratings following a Level I restructuring, versus 0.13 and 0.06 points, respectively, for Level II. These differences indicate that deeper restructurings may generate stronger corrective effects, possibly because they mobilize greater managerial attention and resource reallocation.

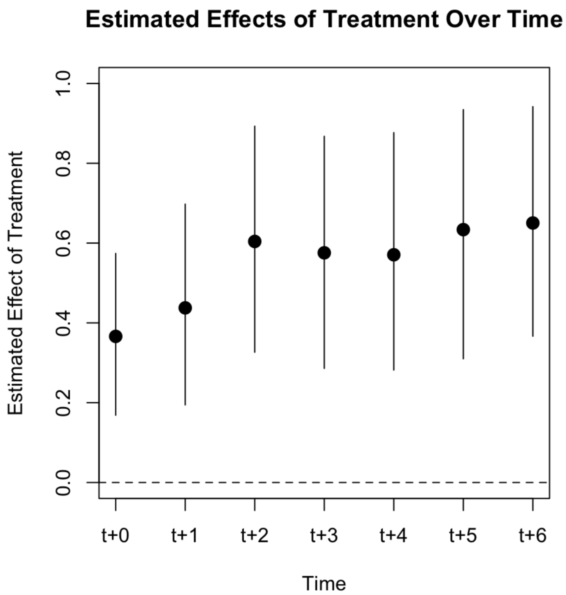
The dynamic results in Figures 8 and 9, estimated using PanelMatch, reinforce this pattern. Both restructuring types yield sustained improvements in ratings over subsequent ISR periods, but the magnitude is consistently higher for Level I cases. Together, these findings underscore the heterogeneity of adaptive responses: while most restructurings help projects recover, more comprehensive adjustments deliver larger and more durable gains.

Table 2: Effect of Restructuring Type on ISR Ratings: Two-Way Fixed Effects Estimates

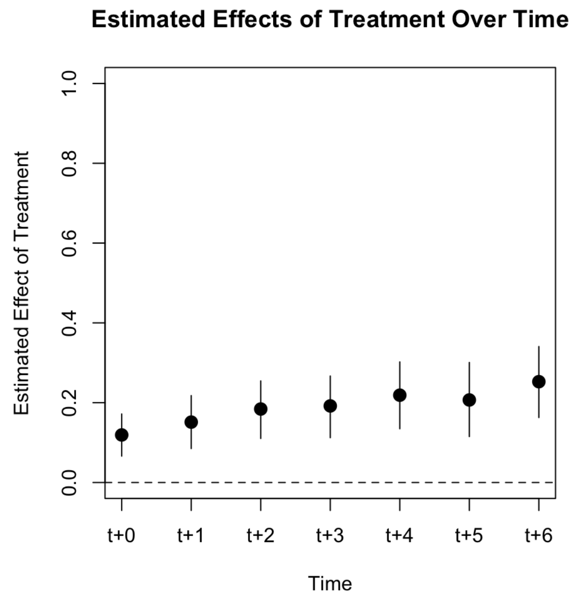
	IP	PDO
Level I	0.293***	0.209***
	(0.056)	(0.059)
Level II	0.132***	0.060***
	(0.016)	(0.015)
Observations	52069	57160
R^2	0.450	0.475
Adjusted R^2	0.383	0.412

Note: Clustered standard errors in parentheses. *p<0.1; **p<0.05; ***p<0.01.

Figure 8: Dynamic Effects of Restructuring on IP Ratings: PanelMatch Estimates

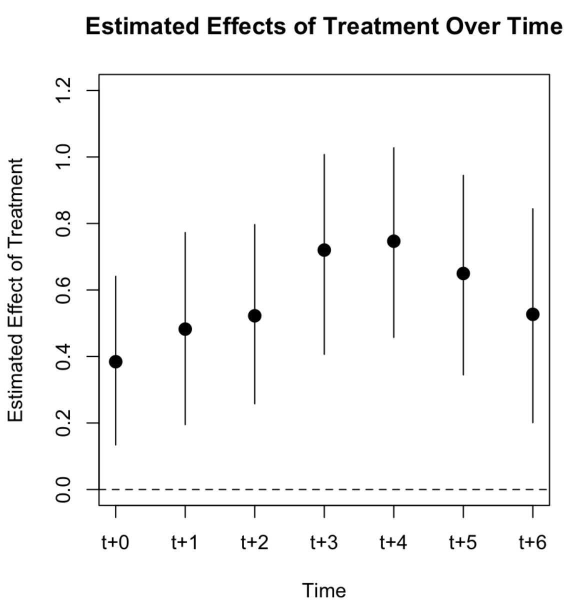


(a) Level I

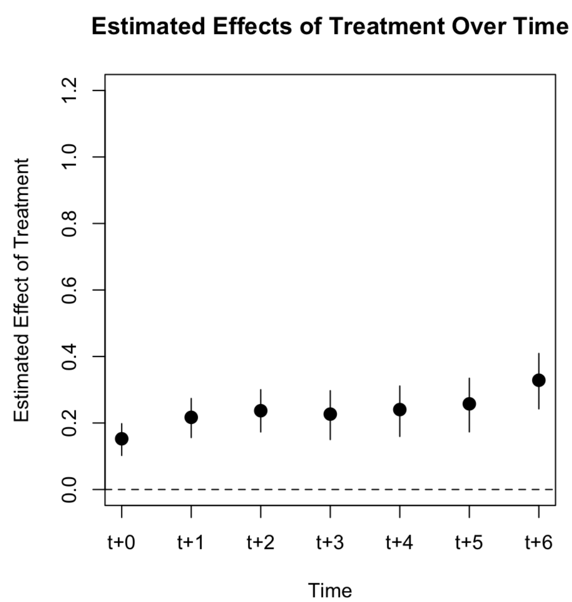


(b) Level II

Figure 9: Dynamic Effects of Restructuring on PDO Ratings: PanelMatch Estimates



(a) Level I



(b) Level II

To examine Hypothesis 3, we extend the baseline specification by differentiating restructurings according to their timing within the project cycle. Specifically, we compute each project's mid-course point as half of its actual implementation duration, measured in dates from approval to closure. We then classify restructurings that occur strictly before this mid-course threshold as early restructurings, and those that occur at or after this threshold as late restructurings.

Using this definition, we construct two mutually exclusive, time-varying treatment indicators. The first, EarlyRestructured, equals one if project i had experienced at least one early restructuring by the time of ISR report t, and zero otherwise. The second, LateRestructured, is defined analogously for restructurings that occur at or after the mid-course point. Formally, both variables follow the same absorbing-state logic as our baseline restructuring indicator: once a project transitions into the "treated" state (early or late), the corresponding indicator remains equal to one for all subsequent ISR reports. Projects that never undergo restructuring remain coded as zero throughout.

Figures 10 and 11 present dynamic treatment effect estimates disaggregated by restructuring timing. Several findings stand out.

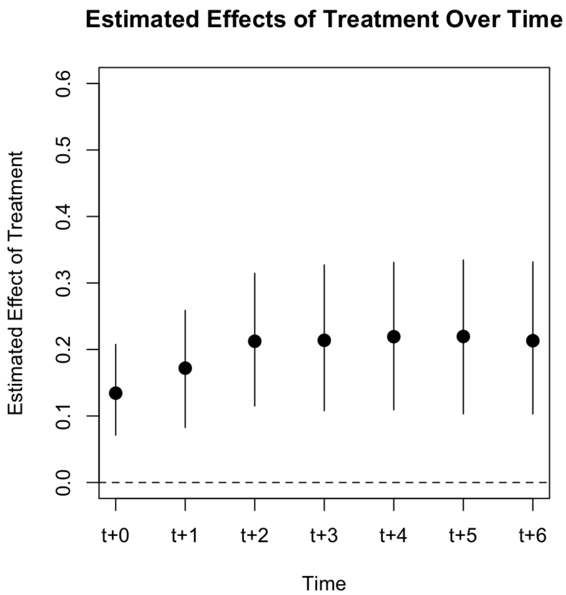
First, both early and late restructurings are followed by positive and statistically significant improvements in ISR ratings. Importantly, neither type of restructuring generates negative or transitory effects; across both IP and PDO, point estimates are consistently above zero.

Second, the trajectory of early restructurings reflects durability rather than magnitude. For IP, early restructurings raise ratings by about 0.13 points at onset ($t+0$), rising modestly to around 0.21 points by lead 4 and holding steady through lead 6. PDO exhibits a similar profile, with effects of 0.20–0.30 points sustained over multiple cycles. This pattern is consistent with the theoretical expectation that projects restructured while implementation structures remain flexible can lock in improvements that persist over successive ISR updates. The fact that these gains remain stable over several years suggests that early restructurings recalibrate project trajectories in durable ways.

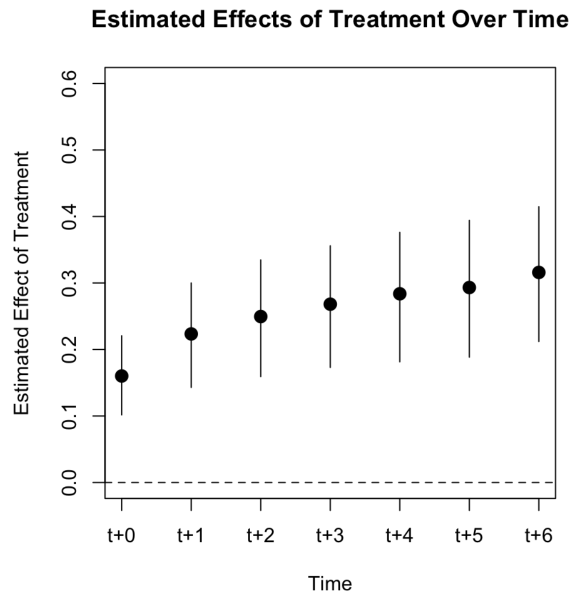
Third, late restructurings also deliver substantial improvements, in some horizons of even larger point estimates than early restructurings – approaching 0.30–0.35 ISR points. These gains are noteworthy because they occur despite compressed timelines and hardened contractual and political relationships. However, the nature of late restructuring is inherently constrained: projects often have only one or two ISR cycles remaining before closure, leaving limited opportunities for improvements to accumulate or become fully institutionalized. Moreover, confidence intervals widen more quickly in the late sample, reflecting both reduced sample size and greater heterogeneity in outcomes.

The substantive difference between early and late restructurings is thus not that one "works" and the other does not, but that early restructurings extend the horizon over which improvements can be realized and reinforced. Timing matters because the opportunity for ratings to respond is a function of how much of the project's lifecycle remains, not because late restructurings are ineffective. In practical terms, this means that task teams and managers who restructure early can lock in durable gains and lessons that accumulate over time, while those who restructure late can still secure meaningful improvements, but only briefly and under tighter constraints. Early restructurings leverage the full project lifecycle to consolidate improvements, whereas late restructurings are inherently bounded by time. Understanding this distinction is critical for how international organizations manage long-term projects in uncertain environments.

Figure 10: Dynamic Effects of Early Restructuring on ISR Ratings: PanelMatch Estimates

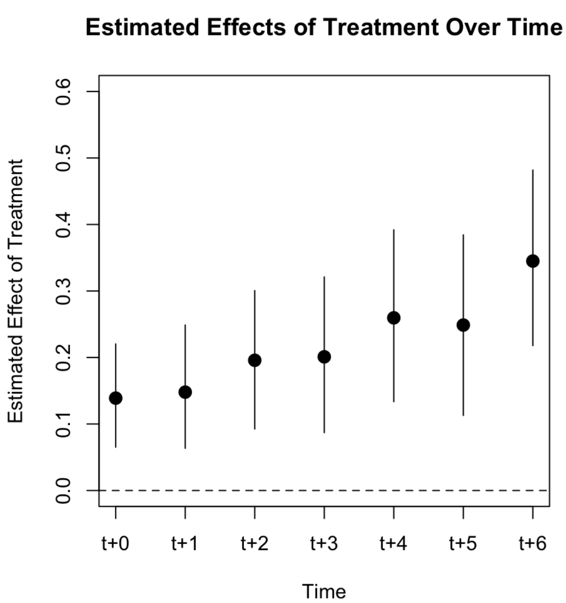


(a) IP

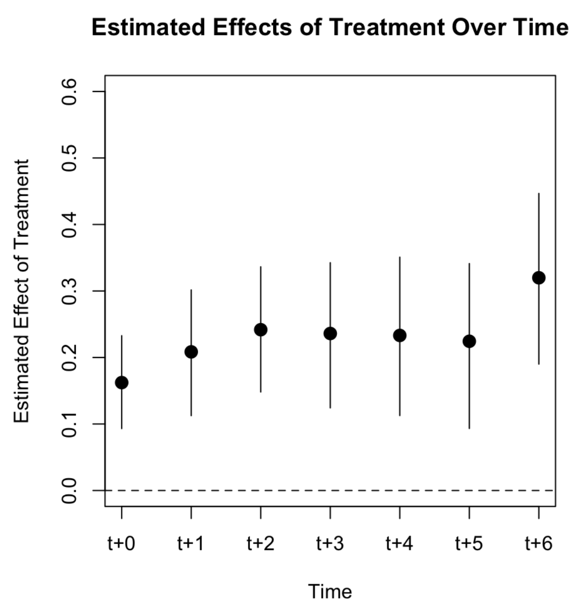


(b) PDO

Figure 11: Dynamic Effects of Late Restructuring on ISR Ratings: PanelMatch Estimates



(a) IP



(b) PDO

7 Robustness Checks

A potential concern with our specification is the treatment of the outcome variable. In the main analysis, we treated the ISR rating as a cardinal variable ranging from 1 ("Highly Unsatisfactory") to 6 ("Highly Satisfactory"). This approach follows common practice in empirical work on World Bank project performance, where the numerical scale facilitates a straightforward interpretation of within-project changes over time. Nevertheless, the ISR scale is formally ordinal: the conceptual distance between adjacent categories is not uniform, and ratings below 4 carry a disproportionate negative signal in operational contexts. Because the distribution is right-skewed – with few projects rated below "Moderately Satisfactory" – assuming equal intervals between categories could distort the estimated effect sizes.

To address this concern, we recoded the outcome variable into a binary measure distinguishing unsatisfactory from satisfactory outcomes. Ratings of 1–3 ("Highly Unsatisfactory," "Unsatisfactory," "Moderately Unsatisfactory") were coded 0, and ratings of 4–6 ("Moderately Satisfactory," "Satisfactory," "Highly Satisfactory") were coded 1. This dichotomization mirrors the Bank's internal performance classification, where ratings below 4 trigger managerial attention and potential remedial actions.

We then re-estimated all models using a fixed-effects logit specification of the form:

$$\text{logit}(\Pr[\text{Satisfactory}]_{i,t} = 1) = \beta_0 + \beta_1 \text{Restructured}_{i,t} + \alpha_i + \delta_f + \epsilon_{i,t} \tag{2}$$

The logit model is appropriate here because it treats the outcome as categorical and constrains predicted probabilities to the $[0, 1]$ range, avoiding the linear-probability extrapolation inherent in OLS. The fixed-effects structure ensures that identification continues to rely on within-project changes over time, comparable to the baseline two-way fixed-effects (TWFE) model. This provides a nonlinear analogue to our main specification that respects the ordinal nature of the outcome variable while maintaining interpretability in terms of odds ratios rather than linear changes in the outcome index.

For completeness, we applied the same binary specification to the PanelMatch framework. The treatment indicator and covariate set remained unchanged. Results from both the TWFE-logit and PanelMatch binary estimations are reported in Appendix D. Treating the outcome variable as dichotomous therefore yields inferences consistent with those obtained under the cardinal specification, suggesting that our findings are not sensitive to assumptions about the scale of the ISR ratings.

8 Discussion

The evidence in this paper can be interpreted through the lens of adaptive management (Holling 1978; Walters 1986) and organizational learning (Levitt and March 1998; March 1991) in complex delivery environments. In high-uncertainty settings, the canonical prescription is not to "get the design right" ex ante and then execute mechanically, but to embed structured feedback loops that allow the implementer to update strategy midstream. Restructuring, in the World Bank context, is precisely such a structured feedback loop. It transforms informal problem recognition into a formal, reviewable, and monitorable adaptation. This makes restructuring analytically comparable to what the adaptive management literature calls a learning event: a bounded moment in which managers absorb evidence, revise their theory of change, and reallocate attention and resources accordingly. By showing that ISR ratings improve after such events and improve more durably when they occur early, we provide empirical content to that learning story.

Teams that restructure are not just reacting; they are demonstrating the ability to interpret noisy operational data, translate it into an implementable revision, and internalize that revision as the new baseline against which they will be judged. That capability – taking real-time performance intelligence seriously enough to change course before closure – is a scarce managerial skill in large public programs. They are the institution's real-time record of whether teams can learn under pressure. In that sense, our findings speak beyond the World Bank. Any large organization that finances multi-year, politically exposed programs (for example, sovereign lending institutions, climate funds, vertical health initiatives, even domestic infrastructure agencies) faces the same problem: initial designs get outdated faster than bureaucracies are wired to admit. Our results show that making "design updates under uncertainty" a routinized, reviewable act, rather than an admission of failure, has measurable payoffs in internal performance assessments. That is, restructurings are not exceptions to the model; they are the model of credible delivery in complex systems.

The timing result in this paper is more than a descriptive regularity. It implies that responsiveness has a horizon. Early in the project, governance arrangements, procurement architecture, and stakeholder expectations are still malleable; the Bank and the borrower can still rewrite which ministry does what, re-scope an underperforming subcomponent, or redirect funds toward the binding bottleneck. When restructuring happens in that window, the corrected trajectory has time to compound. ISR ratings absorb that sustained improvement: Implementation Progress moves because upstream delivery constraints have actually been relaxed, and Project Development Objective ratings move because the objective itself now better matches what can plausibly be delivered before closing. By contrast, late restructurings occur under tighter temporal and political constraints. They still improve ratings – often sharply – but they do so in a compressed number of remaining reporting cycles. The adjustment is credible, but it cannot durably propagate because the project is already approaching its legal closing date. In other words, "late" is not ineffective; "late" is bounded.

The distinction between Level I and Level II restructurings reinforces this interpretation. Level I restructurings – those requiring Board approval – tend to be more comprehensive, addressing multiple interlocking constraints at once: revising the development objective, reallocating funds, and redefining institutional responsibilities. They are more costly administratively but yield deeper learning, as they force both teams and management to revisit the project's original theory of change. Level II restructurings, by contrast, typically involve incremental adjustments within the existing design – such as reallocation across components or minor revisions to results indicators. Our findings show that Level I restructurings are associated with larger subsequent improvements in ISR ratings, consistent with the view that meaningful adaptation, not marginal tweaking, drives performance recovery. The evidence thus points to the quality and scope of course correction, not merely its existence or timing, as central to effective adaptive management.

These patterns should not be interpreted as a recommendation to increase the frequency of Level I restructurings. Level I actions are rightly reserved for situations where projects face binding design flaws or require strategic redirection, and they impose higher administrative and transaction costs. Our interpretation is instead that when meaningful course correction is warranted, deeper and more decisive adjustments appear more strongly associated with improved performance. The empirical patterns therefore reflect the informational content of a Level I restructuring – its signal of substantive adaptation – rather than a normative claim about how often such restructurings should occur.

Taken together, our findings suggest that what matters for project performance is the depth and institutionalization of feedback loops. Early, substantive restructurings allow design learning to feed back into implementation, while late or superficial ones rarely reset the system's equilibrium. From a governance perspective, this implies that adaptive capacity is not only temporal but organizational: it depends on whether teams have the autonomy, incentives, and managerial cover to diagnose structural problems early and propose ambitious fixes.

This distinction offers several design lessons for portfolio governance. One implication is to institutionalize structured "Outcome Reflex" checkpoints before the midpoint of implementation – not as compliance policing but as opportunities for reflexive learning. These reviews could coincide with key operational thresholds: (i) 12 months after first disbursement, (ii) before 40% disbursement, and (iii) at the first major procurement milestone for any critical component. A second implication is to embed adaptive performance metrics into managerial incentives: for example, tracking the share of proactive restructurings before the midpoint or the proportion of Level I restructurings yielding positive ISR revisions. Finally, project-level dashboards that integrate disbursement trajectories, ISR risk flags, and milestone slippage could algorithmically surface candidates for early, meaningful course correction. None of these interventions requires new financing; they require normalizing early, substantive adaptation – and rewarding it explicitly. Our empirical findings provide quantitative backing for doing so.

One caveat is that ISR ratings are partially subjective: they are generated by task teams, reviewed by management, and reflect judgments about feasibility and performance. This raises a legitimate question: when we observe ISR ratings improving after restructuring, are we capturing real improvements in delivery, or simply a re-synchronization of expectations with reality? We argue that both channels are present, and that both are operationally meaningful. We label these channels substance improvement and signal improvement. Substance improvement refers to cases where restructuring removes a genuine operational bottleneck – unblocking procurement, reallocating funds toward the binding constraint, clarifying institutional responsibilities, or cancelling a chronically underperforming activity. In those cases, Implementation Progress ratings rise because the project actually starts moving. Signal improvement, by contrast, occurs when the restructuring updates the Results Framework, streamlines indicators, or revises targets so they are logically aligned with what can be delivered in the remaining time and political space. Here, PDO ratings become more favorable not because the underlying development challenge suddenly got easier, but because the stated objective is now calibrated to what is politically and administratively feasible.

Treating signal improvement as "mere cosmetics" would be a mistake. In multi-year sovereign operations, credibility is itself an input into performance. Before restructuring, an operation can be stuck in a credibility trap: the team continues to report against indicators everyone privately knows are obsolete, and ISR ratings drift downward as feasibility collapses. After restructuring, the same operation can report against a sharper, internally coherent objective set that both parties – Bank and borrower – believe they can deliver. That credibility shift matters for two reasons. First, it unlocks managerial attention and resources for the revised objective, rather than wasting supervision bandwidth defending an unrealistic Results Framework. Second, it protects reputational capital: a borrower that sees the Bank as willing to acknowledge and formalize reality has stronger incentives to remain engaged rather than disengage defensively. In this sense, upward movement in ISR ratings after restructuring is not evidence of "leniency." It is evidence that expectations and delivery plans have been brought back into alignment, which is precisely what adaptive management is supposed to do. Our empirical strategy cannot fully decompose substance from signal, but the persistence of gains over multiple ISR cycles, especially after early restructurings, is more consistent with durable realignment than with one-off cosmetic relabeling.

9 Conclusion

We examine how project restructurings shape performance during implementation. Drawing on ISR ratings, we track the immediate and dynamic effects of restructuring across a large sample of World Bank projects. We find that restructurings are consistently followed by improvements in both Implementation Progress (IP) and Project Development Objective (PDO) ratings, and that these gains accumulate over successive reporting cycles. The magnitude and persistence of these effects depend on both timing and scope. Early restructurings yield the most durable improvements, as design corrections have time to compound. Moreover, Level I restructurings – those requiring Board approval and entailing substantive design changes – generate stronger and more sustained rating upgrades than Level II restructurings, which typically involve narrower, procedural adjustments. Even late or limited interventions, however, produce short-term gains, suggesting that adaptation at any stage can partially restore project performance, though its durability is bounded by remaining implementation time.

These findings carry several policy implications. First, while the Bank formally recognizes restructuring as part of the project cycle, in practice it is still often perceived as a remedial step rather than a normal management tool. Our evidence suggests that restructuring should instead be treated as a routine instrument of adaptive management. Strengthening internal incentives that reward timely course correction – rather than penalizing projects for departing from original designs – could raise the overall quality of the portfolio. Second, the results underscore the need for institutionalized early reviews. Embedding structured checkpoints within the first half of project implementation would enable teams to identify design flaws and recalibrate strategies before trajectories harden. Third, to sustain credibility and learning, both donors and client governments should be encouraged – through communication protocols and reporting standards – to view restructurings as evidence of managerial responsiveness. Framing restructurings as proactive adjustments, rather than crisis interventions, would enhance the Bank's legitimacy and its capacity to deliver results under uncertainty.

Several directions for future work follow. One is to explore heterogeneity: whether the impact of restructuring varies across sectors, borrower types, or institutional environments. Another is to examine the organizational drivers of timing – why some task teams restructure early while others delay until problems accumulate. A third avenue is to investigate complementarities, such as how restructuring interacts with supervision quality, leadership continuity, or the use of results frameworks in shaping trajectories. Future work could also assess the incremental effect of having multiple restructurings – whether a second or third restructuring yields diminishing, compounding, or neutral returns. Each of these extensions would help identify not only whether restructuring works, but also when and why it is deployed most effectively.

We view these findings as part of a broader "Outcome Reflex" agenda. In practical terms, the Bank now has the ingredients for early-warning models that predict when and how to recalibrate projects for the largest payoff.

An obvious extension is to train models that estimate (i) the probability that a given operation will require restructuring in the next few ISR periods, and (ii) whether we remain within the window where an early and appropriately scoped restructuring would generate multi-cycle improvements. Embedding that logic into AI-enabled supervision systems would involve three concrete steps.

First, automatic triage: models flag operations whose ISR language shows drift and whose disbursement profile is decoupling from plan, routing them for a diagnostic checkpoint.

Second, timing and degree guidance: rather than waiting for formal downgrades to Moderately Unsatisfactory, managers receive forward-looking alerts such as "if this operation restructures within the next six months, the historical likelihood of persistent ISR improvement is X; if it waits another year, the expected gain is Y but lasts only one ISR cycle," along with indications of whether a moderate or major restructuring historically delivers higher returns under similar conditions.

Third, portfolio learning: by aggregating these signals across regions and Global Departments, senior management can identify not only which projects are at risk, but also which systemic bottlenecks repeatedly push operations toward reactive, late-cycle restructurings.

We would note that AI does not replace managerial judgment; rather, it lowers the cost of acting early and at the right scale. Linking predictive analytics to the governance reforms discussed above would shift the institution from celebrating “good clean-up stories” at closing to institutionalizing early, well-calibrated course correction as standard operating procedure.

Our results underscore that development effectiveness depends not only on strong preparation but also on flexibility and agility during execution. Strengthening the Bank’s ability to learn and adapt in real time – through timely restructurings, robust monitoring, and transparent disclosure – will be essential as projects confront increasingly complex and shock-prone environments.

Policy Implications for TTLs and Development Practitioners

- ***Normalize restructuring.*** Treat restructuring as a routine adaptive tool, not a sign of distress. Early action – before mid-implementation – generates durable ISR improvements.
- ***Act early and decisively.*** Early, substantive restructurings produce stronger and longer-lasting gains than small, late adjustments. Timing and scope work best in tandem.
- ***Institutionalize early checkpoints.*** Embedding structured "Outcome Reflex" reviews before 40% disbursement can surface design flaws before trajectories harden.

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Appendix

The appendix includes the following:

- Appendix A: Summary Statistics
- Appendix B: Current World Bank Restructuring Policies
  - B.1: Investment Project Financing
  - B.2: Program-for-Results Financing
- Appendix C: Institution-Specific Stopword List
- Appendix D: Robustness Check Results

Appendix A Summary Statistics

Table A.1: Summary Statistics

Statistic	Min	Max	Mean	Median	St. Dev.
PDO (1–6 scale)	1	6	4.413	5	0.778
IP (1–6 scale)	1	6	4.257	4	0.799
PDO (binary)	0	1	0.884	1	0.320
IP (binary)	0	1	0.847	1	0.360
(Logged) commitment	2.918	22.357	17.408	17.666	1.766
Restructured	0	1	0.238	0	0.426
Restructured (Level I)	0	1	0.014	0	0.119
Restructured (Level II)	0	1	0.231	0	0.421
Early Restructured	0	1	0.115	0	0.319
Late Restructured	0	1	0.122	0	0.327

Appendix B Current World Bank Restructuring Policies

B.1 Investment Project Financing

- During implementation the Bank, the Borrower, and the member country, as appropriate, may agree to restructure the Project to strengthen its development effectiveness, modify its development objectives, improve Project performance, modify indicators, address risks and problems that have arisen during implementation, make appropriate use of undisbursed proceeds of a Bank Loan, cancel unwithdrawn amounts of a Bank Loan prior to the Loan Closing Date, extend the Closing Date, or otherwise respond to changed circumstances.
- A restructuring involving:
  - (a) a change in safeguard category from a lesser category to a Category A (as defined in OP 4.01 or OP 4.03, as applicable)
  - (b) an extension of the Bank Guarantee Expiration Date
  - (c) reliance on alternative procurement arrangements referred to under Section III.F of the Procurement Policy, or
  - (d) in the case of the MPA Program, (i) a substantive or significant change in the MPA Program's development objective(s), or (ii) the addition of a new borrower(s), is referred to as a Level One restructuring and is submitted for consideration by the Board (or by Management, when the approval authority for the original IPF or MPA Financing, as the case may be, resides with Management).
- A restructuring involving any other modification of the Project is referred to as a Level Two restructuring. Management has the delegated authority to approve Level Two restructurings. Management periodically informs the Board of the Level Two restructurings.

B.2 Program-for-Results Financing

- During the implementation of the PforR Program, and as part of Bank implementation support, the PforR Program may, with the agreement of the Bank and the Borrower, be restructured to strengthen its development impact, modify its development objectives or disbursement-linked indicators, improve PforR Program performance, address risks and problems that have arisen during implementation, make appropriate use of undisbursed financing, cancel unwithdrawn amounts prior to the Closing Date, extend the Closing Date, or otherwise respond to changed circumstances.

- A restructuring involving, under the MPA Program,

  - (i) a substantive or significant change to the MPA Program development objectives
  - (ii) the addition of a new Borrower(s), or
  - (iii) the addition of a new disbursement linked indicator(s), is referred to as a Level One restructuring and is submitted for approval by the Board (or by Management, when the approval authority for the original PforR Financing or MPA Financing, as the case may be, resides with Management).

- A restructuring involving any other modification of the PforR Program is referred to as a Level Two restructuring. The authority to approve Level Two restructuring is delegated by the Board to Management. Management periodically informs the Board of Level Two restructurings.

Appendix C Institution-Specific Stopword List

The stopword list consists of the following terms (in alphabetical order): abbreviation, achievement, achieve, acronym, action, activity, additional, agreement, agency, amount, approval, approve, arrangement, assessment, audit, basic, bank, borrower, capacity, category, change, cilbup, cilbupthe, close, commitment, community, component, complete, country, cost, current, datum, date, dec, december, description, development, dezirohtua, detail, director, disburse, disbursement, document, due, ea, economic, effective, estimate, erusolcsid, expect, finance, financial, framework, fund, fy, government, id, ida, iii, ii, impact, implementation, include, indicator, information, instrument, institutional, investment, iv, june, jun, leader, level, loan, lncrtf, managermanager, management, million, ministry, moderately, month, national, net, objective, organization, original, page, paper, pdo, percent, percentage, period, plan, policy, practice, president, procurement, process, product, program, project, propose, provide, rationale, rate, reallocation, relate, remain, report, request, res, responsible, result, revise, sector, service, sign, status, summary, support, system, table, target, task, team, time, total, trigger, undisbursed, unit, usd, vice, waiver, world.

Appendix D Robustness Check Results

Table D.1: Effect of Restructuring on ISR Ratings: Fixed-effects GLM Estimates

	IP	PDO
Restructured	0.889***	0.793***
	(0.084)	(0.104)
Observations	28,208	24,397
Pseudo R^2	0.14890	0.15462
BIC	53,259.9	45,817.1

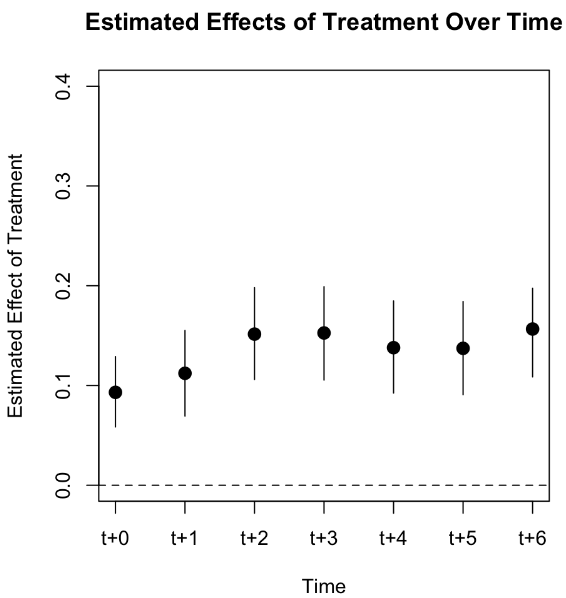
Note: Clustered standard errors in parentheses. *p<0.1; **p<0.05; ***p<0.01.

Table D.2: Effect of Restructuring Type on ISR Ratings: Fixed-effects GLM Estimates

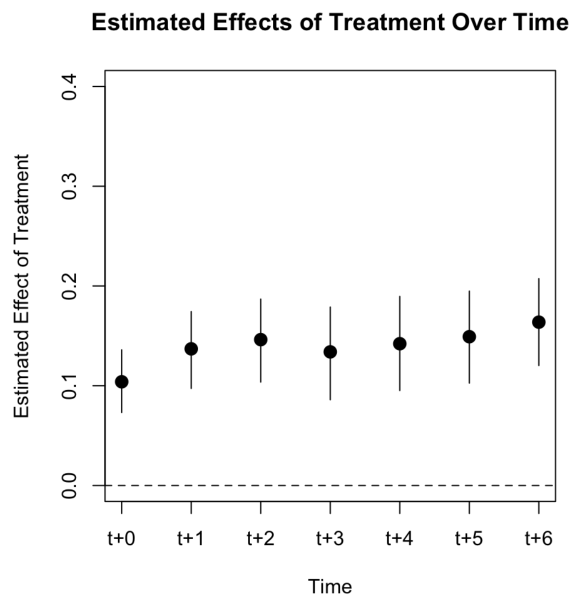
	IP	PDO
Level I	1.40***	1.47***
	(0.217)	(0.229)
Level II	0.860***	0.782***
	(0.085)	(0.105)
Observations	28,208	24,397
Pseudo R^2	0.15075	0.15718
BIC	53,208.9	45,755.1

Note: Clustered standard errors in parentheses. *p<0.1; **p<0.05; ***p<0.01.

Figure D.1: Dynamic Effects of Restructuring on ISR Ratings: PanelMatch Estimates

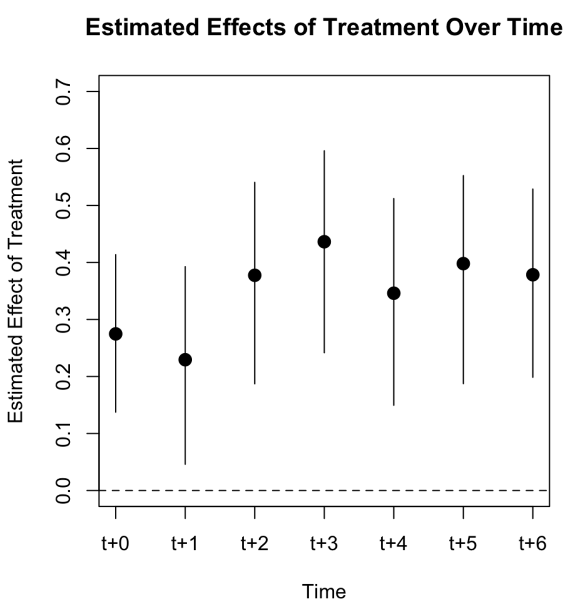


(a) IP

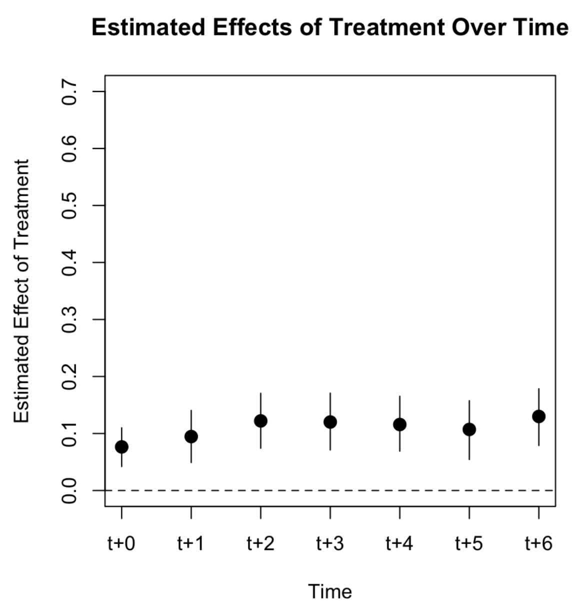


(b) PDO

Figure D.2: Dynamic Effects of Restructuring on IP Ratings: PanelMatch Estimates

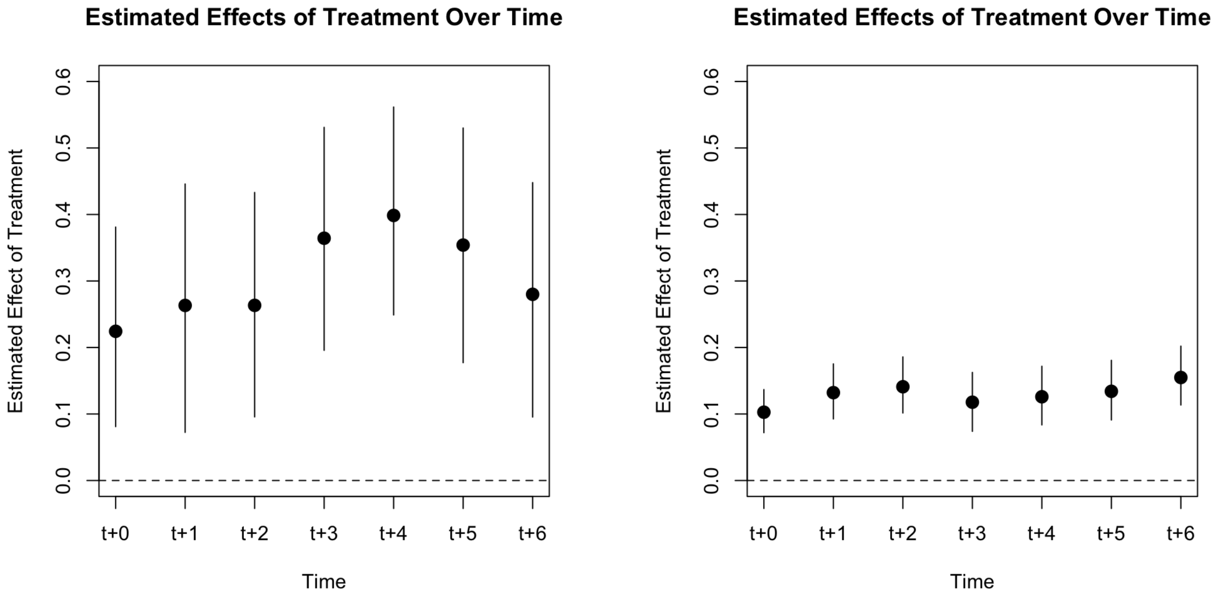


(a) Level I



(b) Level II

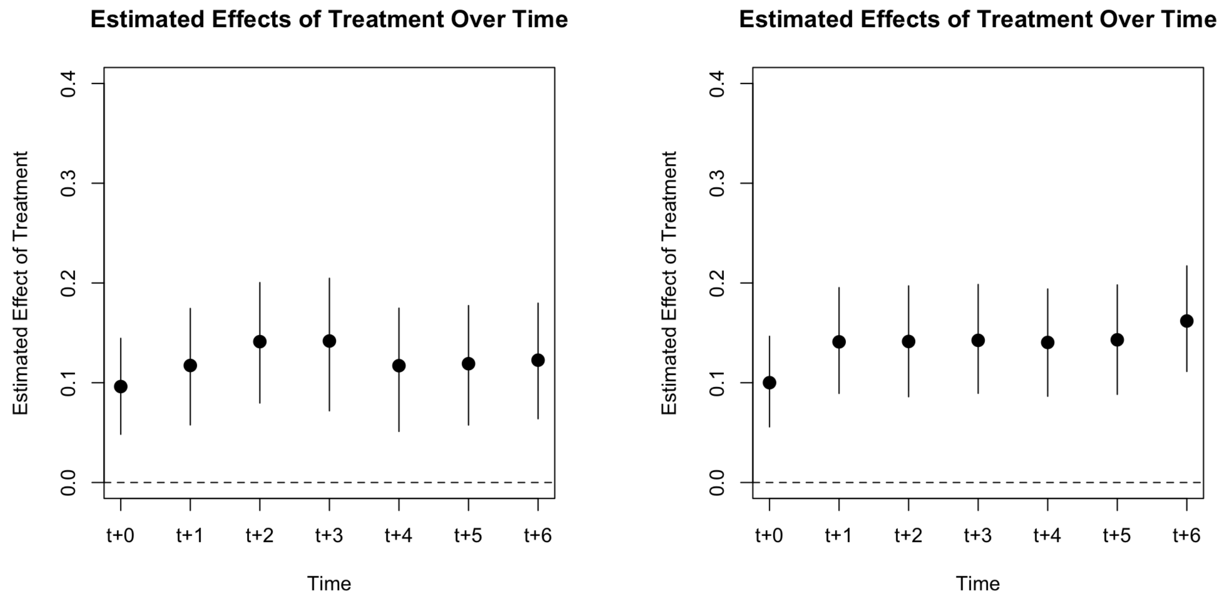
Figure D.3: Dynamic Effects of Restructuring on PDO Ratings: PanelMatch Estimates



(a) Level I

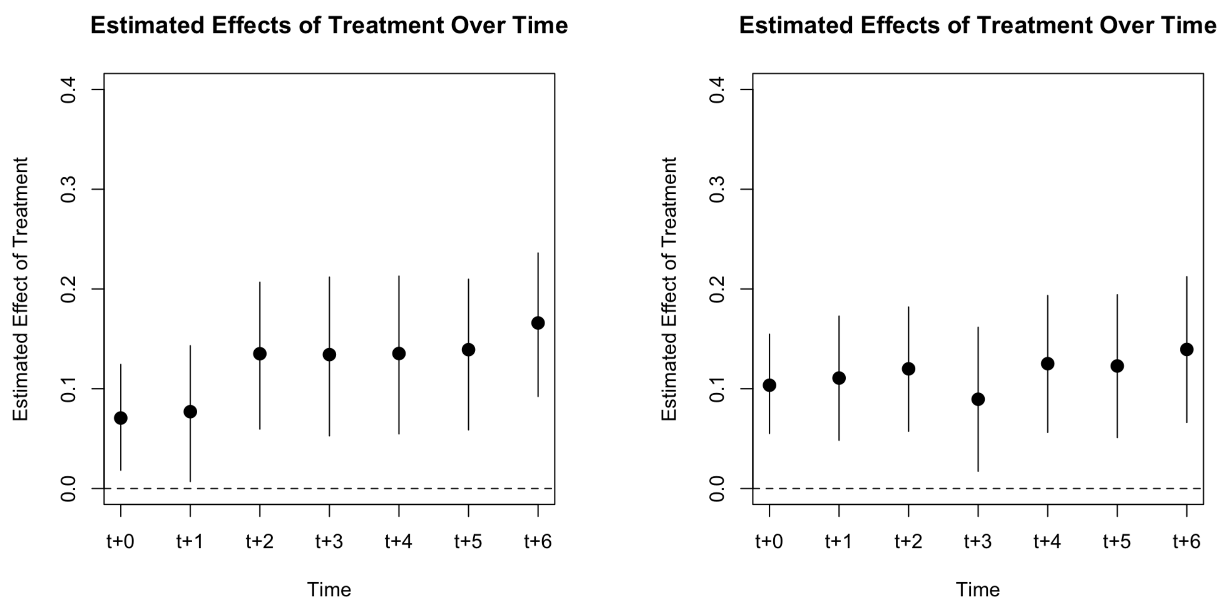
(b) Level II

Figure D.4: Dynamic Effects of Early Restructuring on ISR Ratings: PanelMatch Estimates



(a) IP (b) PDO

Figure D.5: Dynamic Effects of Late Restructuring on ISR Ratings: PanelMatch Estimates



(a) IP (b) PDO